Deep Learning-Based Intrusion Detection System for Smart Home IoT with Reduced False Positives

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1. Introduction

Smart homes are becoming very common today. Many houses now use smart devices such as smart lights, smart TVs, smart cameras, smart locks, voice assistants, and smart thermostats. All these devices are connected to the internet through Wi-Fi. This makes life easier and more comfortable. However, it also creates serious security risks.

Because all devices are connected to the internet, hackers can try to attack the network. They may try to steal data, take control of smart cameras, disable security systems, or launch denial-of-service attacks. Many IoT devices have weak security, which makes them easy targets. Therefore, protecting smart home networks is very important.

One of the main ways to protect networks is by using an Intrusion Detection System (IDS). An IDS monitors network traffic and detects suspicious or malicious activities. Recently, deep learning techniques have been used to improve intrusion detection performance.

This research focuses on developing a deep learning-based intrusion detection system specifically for smart home IoT environments, with special attention to reducing false positive alarms and improving practical deployment.

2. Base Paper

The base paper for this research is:

"A Hybrid Deep Learning Approach for Intrusion Detection in IoT Networks."

In this paper, the authors propose a hybrid deep learning model for detecting attacks in IoT networks. A hybrid model means that it combines more than one deep learning technique, such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), including LSTM or GRU.

The main contributions of the base paper are:

- It uses deep learning models to analyze IoT network traffic.
- It combines multiple models to improve detection accuracy.
- It evaluates performance using standard IoT intrusion datasets.
- It reports high accuracy and low false positive rate.

The hybrid model performs very well in controlled experimental conditions. It achieves strong results in terms of accuracy, precision, recall, and F1-score. This shows that deep learning is effective for detecting intrusions in IoT networks.

However, the model is complex and computationally heavy due to the hybrid architecture. It mainly focuses on maximizing accuracy rather than optimizing the system for real smart home deployment.

3. Problem Statement

Although existing research, including the base paper, shows high detection accuracy, there are still practical challenges:

1. Many systems are tested only in controlled laboratory environments.

2. Hybrid deep learning models can be computationally expensive.
3. Smart home routers and IoT gateways have limited processing power.
4. Even small false positive rates can cause frequent unnecessary alarms in real homes.

   In real-life smart homes, too many false alarms can reduce user trust in the system. If the system keeps generating alerts for normal activities, users may start ignoring warnings.

   Therefore, there is a need to design a lightweight and efficient intrusion detection system that:

- Maintains good detection performance.
- Reduces false positive rate.
- Can be deployed in resource-constrained smart home environments.

4. Proposed Research Idea

This research proposes a Deep Learning-Based Intrusion Detection System for Smart Home IoT that focuses on reducing false positives while maintaining strong detection performance.

Instead of using a complex hybrid deep learning architecture like in the base paper, this research will:

- Design a simpler and lightweight deep learning model (such as a single CNN or LSTM model).
- Optimize model parameters for smart home environments.
- Apply threshold tuning to reduce unnecessary alerts.
- Evaluate performance specifically in the context of smart home traffic.

The system will work as follows:

1. Network traffic from smart home devices will be collected.
2. The data will be preprocessed and normalized.
3. The deep learning model will be trained to learn normal and malicious patterns.
4. During real-time operation, the model will classify incoming traffic as normal or attack.
5. If suspicious activity is detected, an alert will be generated.

Special focus will be placed on minimizing false positive alarms without significantly reducing detection capability

5. Research Gap

The base paper focuses mainly on achieving high accuracy using a hybrid deep learning approach. However, it does not specifically address:

- Deployment in smart home environments with limited resources.
- Model complexity and computational cost.
- Practical reduction of false alarms in real-world usage.

This research aims to fill that gap by focusing on lightweight model design and reduction of false positive rates, making the system more practical for everyday smart home use.

6. Expected Contributions

The expected contributions of this research are:

1. A lightweight deep learning model suitable for smart home IoT environments.
2. Improved balance between detection accuracy and false positive rate.

3. Practical evaluation focused on smart home deployment.
4. Better usability and trustworthiness due to fewer unnecessary alerts.

7. Conclusion

Smart homes are becoming increasingly common, but they are also vulnerable to cyber attacks due to the large number of connected IoT devices. Intrusion detection systems based on deep learning have shown strong performance in detecting attacks.

The base paper, “A Hybrid Deep Learning Approach for Intrusion Detection in IoT Networks,” demonstrates the effectiveness of hybrid deep learning models for IoT intrusion detection. However, such models can be complex and computationally heavy.

This research proposes a simplified and optimized deep learning-based intrusion detection system specifically designed for smart home environments. The main goal is to reduce false positive alarms while maintaining strong detection performance.

By focusing on practical deployment and reliability, this work aims to contribute to safer and more efficient smart home security systems.